

Ishita Chauhan

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Education

University of Massachusetts, Amherst

Amherst, MA, USA

Master of Science in Computer Science, GPA: 3.76

May 2026

- Coursework: Advanced Natural Language Processing, Machine Learning, Probabilistic Graphical Models, Methods of Applied Statistics, Data Science Fundamentals, Systems for Data Science, Theory and Practice of Software Engineering, Advanced Algorithms, Research Methods in Empirical Computer Science

Vellore Institute of Technology

Vellore, India

Bachelors in Technology (Major: Computer Science), CGPA: 8.73

July 2024

- Coursework: Natural Language Processing, Image Processing, Machine Learning, Data Structures & Algorithms, Operating System, Theory of Computation, Network and Communication

Experience

UMass Amherst (SIMS Lab)

Amherst, MA

Student Researcher

Sep 2025 – Present

- Operationalized a multi-stage annotation taxonomy for election-related social media discourse (relevancy → target entity → claim type) into an automated Large Language Model (LLM)-driven classification pipeline; evaluated zero-shot, few-shot, and Chain-of-Thought (CoT) prompting strategies across all stages.
- Achieved 92–98% accuracy on relevancy detection, 66–77% on target category classification, and 66–74% on claim type classification; few-shot CoT prompting consistently outperformed other strategies.
- Curated and processed 9,995 posts from X (Twitter) and Truth Social via campaign-period filtering and keyword/Optical Character Recognition (OCR)-based identification of posts containing numeric opinion estimates.

Affine Technologies

Bangalore, India

Machine Learning Intern

Jun 2024 – Jul 2024

- Designed and developed an in-house collaborative filtering recommendation pipeline using KNNWithMeans, replacing a third-party dependency while maintaining comparable accuracy.
- Conducted large-scale Exploratory Data Analysis (EDA) on service usage data to identify regional demand patterns and feature signals for recommendation modeling.

Lifesight Technologies

Bangalore, India

Software Engineering Intern

May 2023 – Jun 2023

- Built the frontend for the Marketing Mix Modeling (MMM) module dashboard in a Business-to-Business Software-as-a-Service (B2B SaaS) platform using Next.js, TypeScript, Sass; integrated Databricks-backed API responses into customer-facing data visualizations.
- Developed interactive geo-visualization features using Mapbox GL to render live campaign analytics data with real-time marker updates and attribute overlays, enabling data-driven decision-making for clients.

Projects

Emotionally Intelligent Therapy Assistant (Hybrid Cache-Augmented Generation (CAG) + Retrieval-Augmented Generation (RAG))

Python, GPT-4o-mini, LLaMA 3.2 1B, ChromaDB

- Built a therapy chatbot using hybrid CAG+RAG memory for long-horizon context retention and safety grounding.
- Implemented session-memory retrieval with structured summaries and top- k recall ($k = 2$) via ChromaDB and a context classifier.
- Fine-tuned GPT-4o-mini on 1,200 multi-turn conversations (960/240 train-eval) with reproducible preprocessing.
- Evaluated turn/conversation quality using embedding cosine similarity (all-MiniLM-L6-v2); improved avg similarity 63.07% to 67.51%.

Real-Time Traffic Congestion Analytics Pipeline

Python, PySpark, Kafka, Parquet

- Built an end-to-end PySpark Structured Streaming pipeline ingesting the Caltrans PeMS District 7 dataset; processed 5-second tumbling windows with a 20-second watermark to compute per-region avg speed, vehicle volume, and congestion index.
- Simulated a live feed by periodically injecting chunk files into the stream; configured Parquet-backed append-mode output with fault-tolerant checkpointing for schema-preserving persistence.
- Instrumented a custom foreachBatch metrics sink to log batch latency and throughput; extended pipeline to Kafka-backed ingestion for decoupled, scalable stream processing.

UMass Hangout System

Spring Boot, React, MySQL, Elasticsearch, WebSockets, Docker

- Built a full-stack hangout platform for creating/joining groups and coordinating meetups across campus.
- Added real-time chat and updates via WebSockets; integrated Elasticsearch for scalable search over groups/messages.
- Containerized and deployed multi-service architecture using Docker (React UI, Spring Boot API, MySQL, Elasticsearch).

Technical Skills

Languages: Python, Java, SQL, JavaScript/TypeScript, HTML/CSS

ML/Natural Language Processing (NLP): PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, Retrieval-Augmented Generation (RAG), Cache-Augmented Generation (CAG), FAISS, Embeddings, Fine-tuning, Model Evaluation

Frameworks: FastAPI, Spring Boot, React, Next.js, Streamlit

Databases/Search: PostgreSQL, MySQL, MongoDB, Firebase, ChromaDB, Elasticsearch

Tools/Infra: Git, Docker, Linux, Jupyter, PySpark, Kafka, Parquet